Security and privacy review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete asset, actor and return-path record.

Keep data-flow map, trusted permission, denied traces, minimized fields and remaining review gaps together.

Keep one habit: Recall → W1–W3/W5 → independent export case → privacy path project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.
- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.
- Security and **privacy review** checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.
- For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

Performance and cost experiments

> **My remembered idea, example and correction**
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Human escalation workflows

> **My remembered idea, example and correction**
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Release and rollback

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Security/privacy review maps assets, actors, actions, data flows and threats. Main G1 guide has an obvious fictional restricted marker; teacher-1 may read, learner-1 may not. A direct denial is insufficient if a query-only student-accessible cache stores/returns the guide. Both repaired direct/cache paths use trusted actor permission before lookup/context; learner denies with empty read trace, teacher allows.

  Request schema is exactly resource_id; role claims reject, unknown actors deny. Separate N1 note text belongs to learner-1 and assigned teacher-1 only; teacher-2 denies. Picture analytics needs revision counts, not private notes: two allowed revision-kind events yield revision_count 2, and a full-note event rejects. No note text or learner IDs are sent in that packet, yet tiny-group counts may remain identifying.

  Check linkability, readers and retention separately. Source instructions cannot grant authority, and happy-path checks do not test all misuse. Code uses copied fictional records/markers and no real secrets/accounts. It does not test all injection, retention or unlisted routes and provides no complete security/anonymity guarantee.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Request schema is exactly resource_id; role claims reject, unknown actors deny. Separate N1 note text belongs to learner-1 and assigned teacher-1 only; teacher-2 denies. Picture analytics needs revision counts, not private notes: two allowed revision-kind events yield revision_count 2, and a full-note event rejects. No note text or learner IDs are sent in that packet, yet tiny-group counts may remain identifying.

  Check linkability, readers and retention separately. Source instructions cannot grant authority, and happy-path checks do not test all misuse. Code uses copied fictional records/markers and no real secrets/accounts. It does not test all injection, retention or unlisted routes and provides no complete security/anonymity guarantee.

- **Output** denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Check linkability, readers and retention separately. Source instructions cannot grant authority, and happy-path checks do not test all misuse. Code uses copied fictional records/markers and no real secrets/accounts. It does not test all injection, retention or unlisted routes and provides no complete security/anonymity guarantee.

- A report omits this qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Security/privacy review maps assets, actors, actions, data flows and threats. Main G1 guide has an obvious fictional restricted marker; teacher-1 may read, learner-1 may not. A direct denial is insufficient if a query-only student-accessible cache stores/returns the guide. Both repaired direct/cache paths use trusted actor permission before lookup/context; learner denies with empty read trace, teacher allows.

  Request schema is exactly resource_id; role claims reject, unknown actors deny. Separate N1 note text belongs to learner-1 and assigned teacher-1 only; teacher-2 denies. Picture analytics needs revision counts, not private notes: two allowed revision-kind events yield revision_count 2, and a full-note event rejects. No note text or learner IDs are sent in that packet, yet tiny-group counts may remain identifying.

  Check linkability, readers and retention separately. Source instructions cannot grant authority, and happy-path checks do not test all misuse. Code uses copied fictional records/markers and no real secrets/accounts. It does not test all injection, retention or unlisted routes and provides no complete security/anonymity guarantee.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled security and **privacy review evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of security and **privacy review**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Security and **privacy review**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Security and **privacy review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only.
- Only the learner and assigned teacher may read note text. The dashboard is authorised for aggregate counts, not notes.
- Trace private note content and analytics counts to their permitted readers. Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only.
- Which labelled step matches this detail? Send revision counts without copying private notes.
- 1. Map content path
- 2. Check readers
- 3. Minimise analytics

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Minimise analytics Send revision counts without copying private notes. A **privacy review** follows data and authority through components.

A useful dashboard does not automatically need every underlying private detail.

Changed-case question and optional hint

- Changed-case practice: The dashboard starts storing full notes. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Flag the unapproved content path and remove it from the design.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Flag the unapproved content path and remove it from the design. Its count-only purpose does not grant note-reading permission.

Repair and retain the record

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Review Release and rollback (ai-6/24). Goal: Explain the **mechanism** and terminology of release and rollback. Short practice: Compute A's and B's correctness percentages.

Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

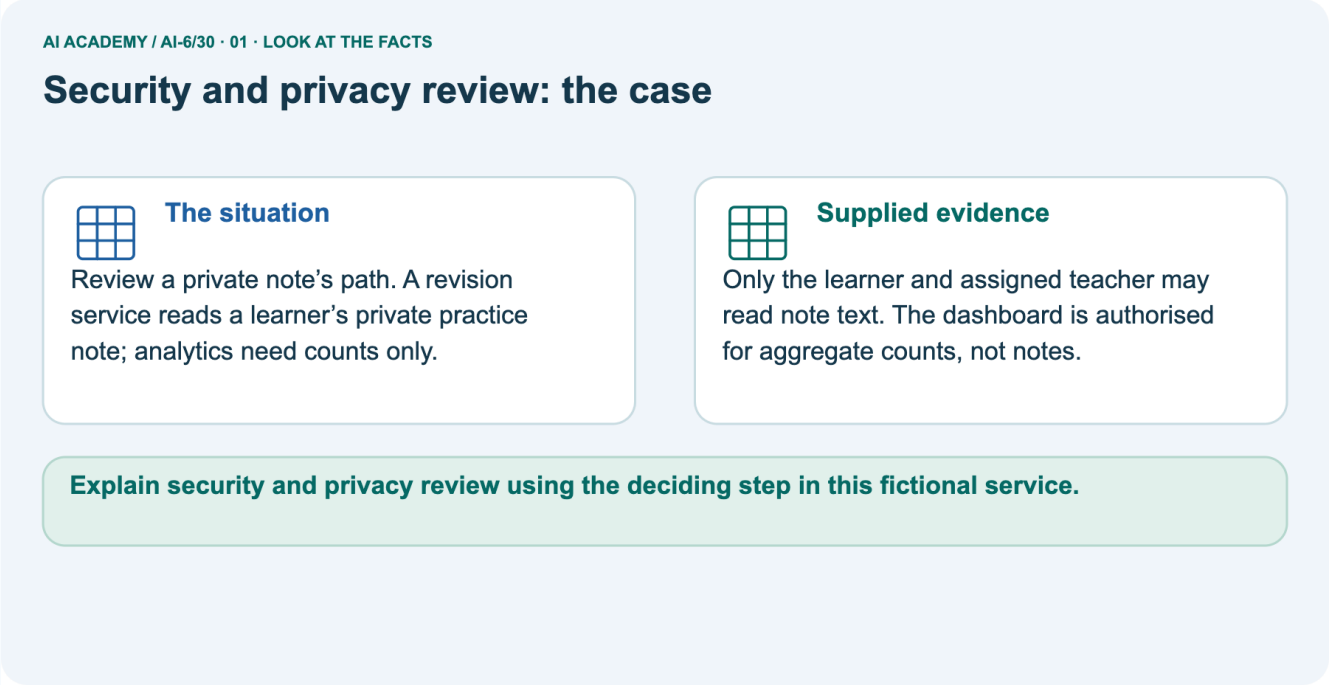
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Security and privacy review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

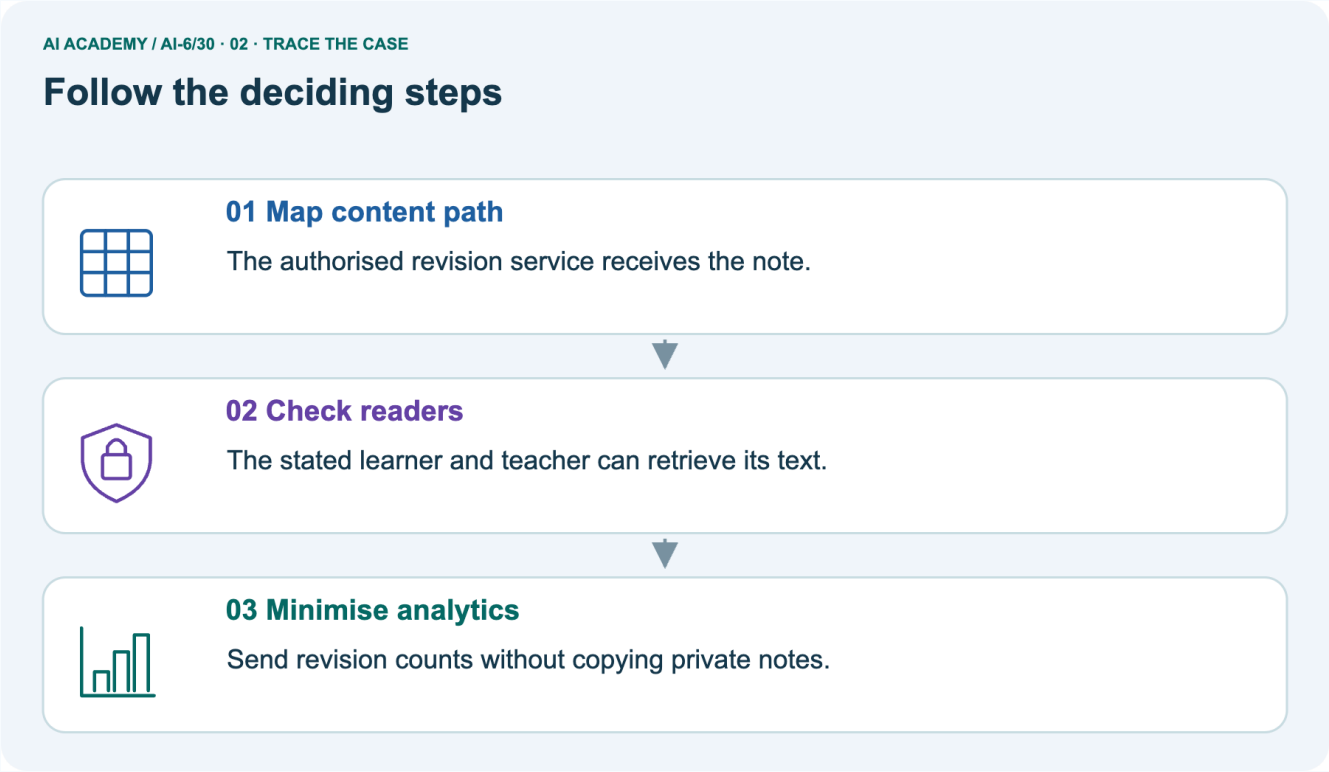
Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only. Supplied facts: Only the learner and assigned teacher may read note text.

The dashboard is authorised for aggregate counts, not notes.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



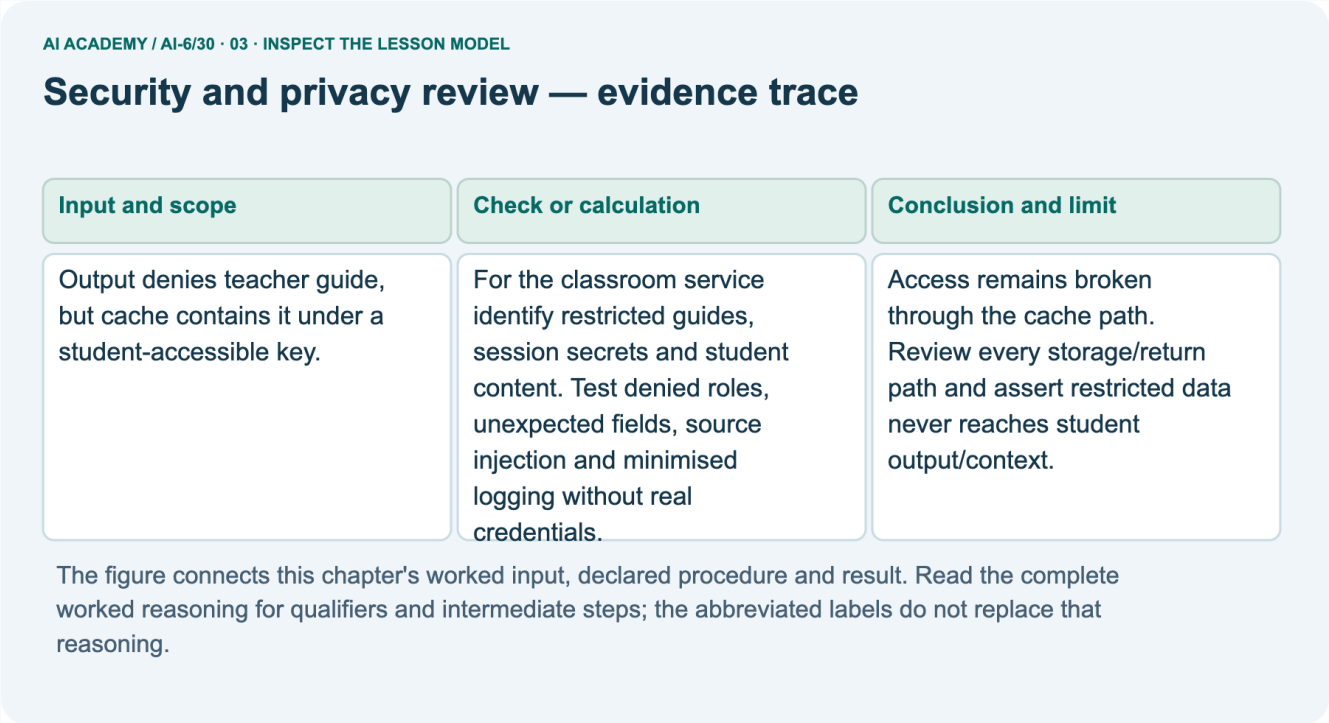
Follow the labelled picture with the matching explanation. Open the vector diagram

Map content path: The authorised revision service receives the note. → Check readers: The stated learner and teacher can retrieve its text. → Minimise analytics: Send revision counts without copying private notes.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Security and privacy review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Security and **privacy review** — **evidence** trace. **Output** denies teacher guide, but cache contains it under a student-accessible key. | For the classroom service identify restricted guides, session secrets and student content.

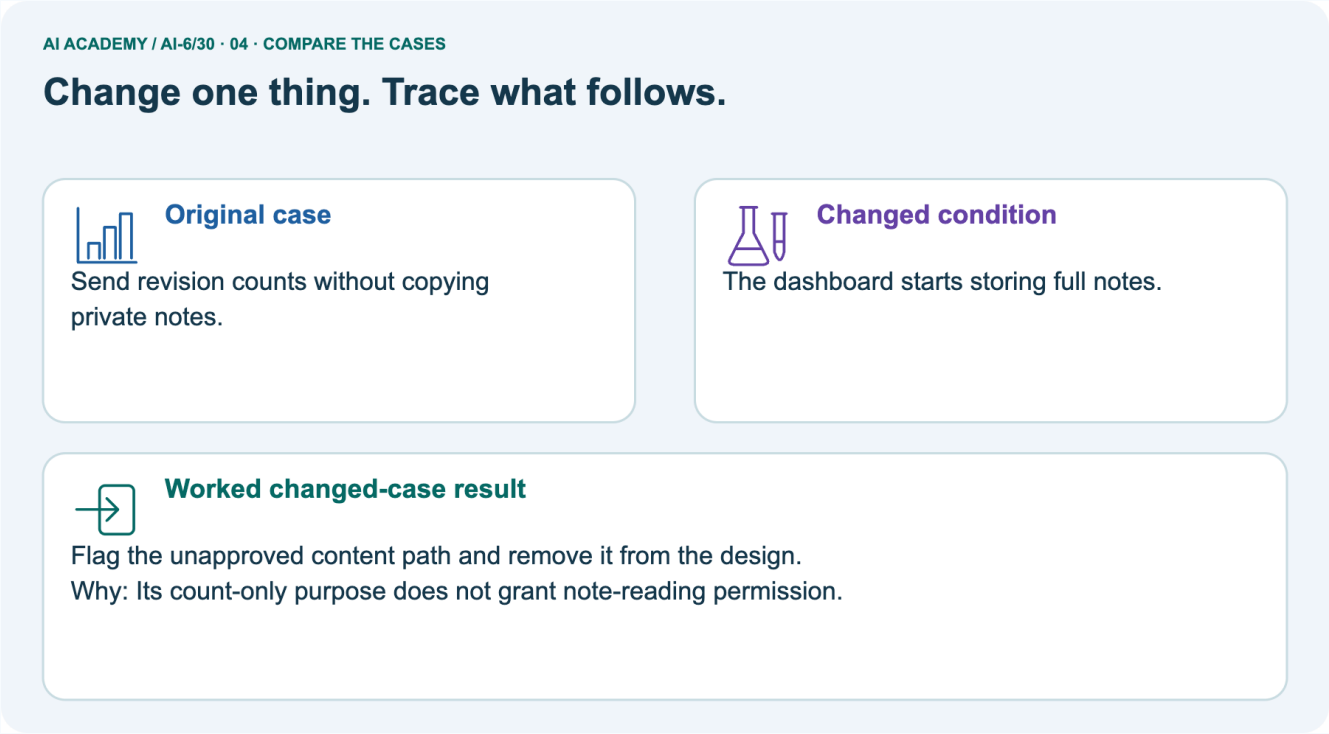
Test denied roles, unexpected fields, source injection and minimised logging without real credentials. | Access remains broken through the cache path.

Review every storage/return path and assert restricted data never reaches student **output**/context.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

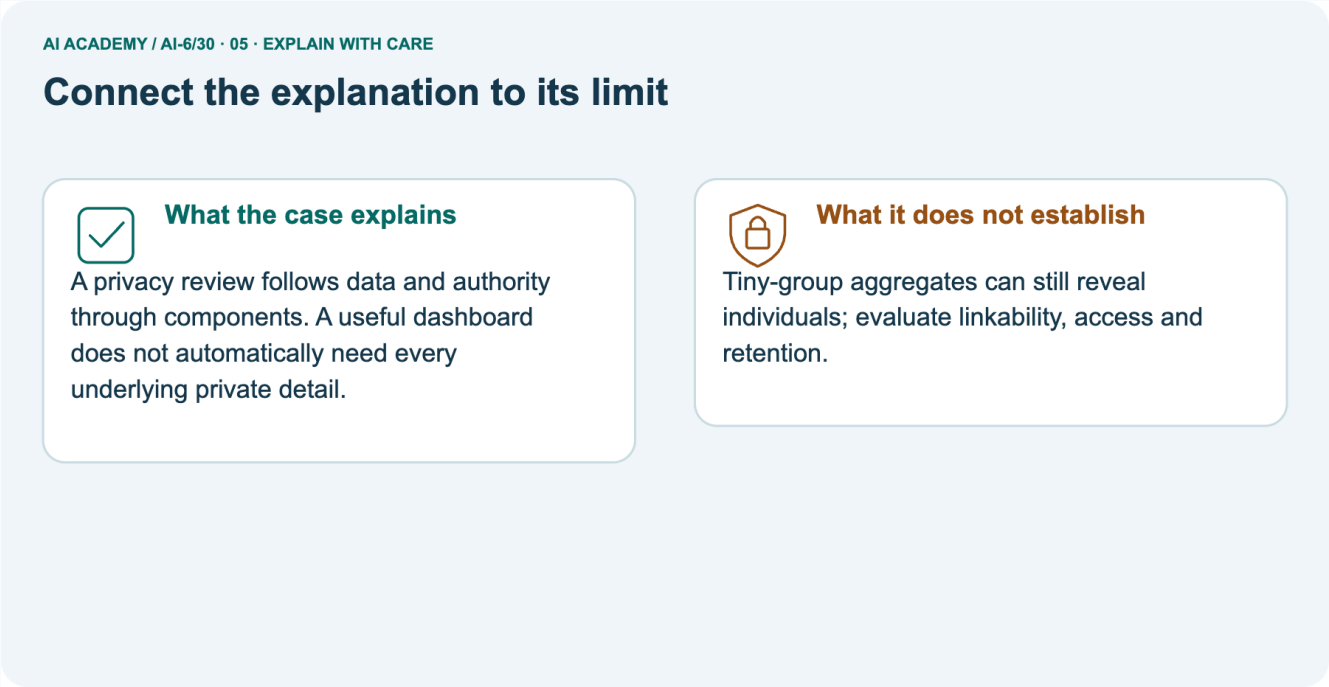
Original case: Send revision counts without copying private notes. Changed condition: The dashboard starts storing full notes. Expected result: Flag the unapproved content path and remove it from the design.

Reason: Its count-only purpose does not grant note-reading permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

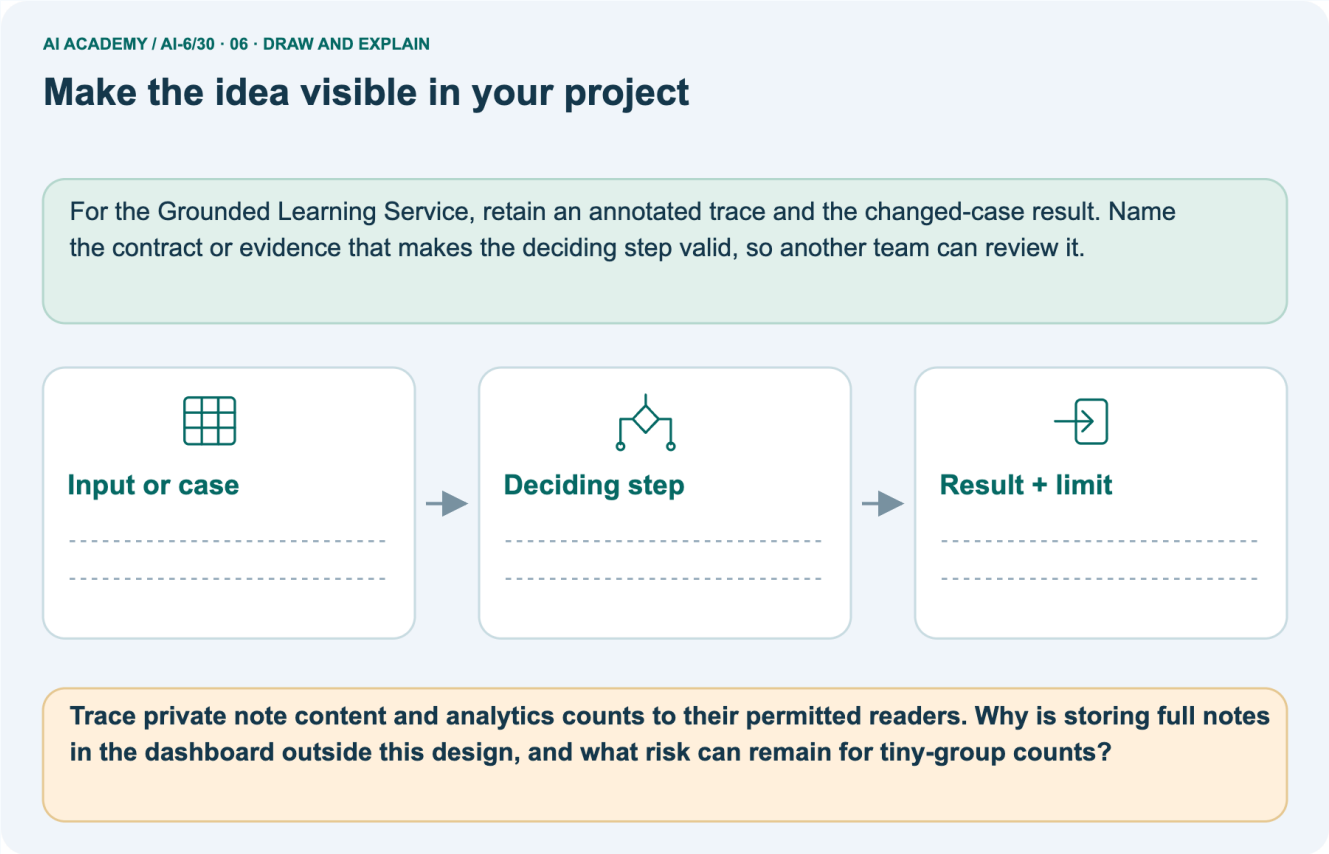
A **privacy review** follows data and authority through components. A useful dashboard does not automatically need every underlying private detail.

Limit: Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Security and privacy review: the case

02. Follow the deciding steps

03. Security and privacy review — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.